

**Remark:** The condition in (5) is best possible. For instance, there is a tripartite graph  $G$  with densities  $\alpha = 1 - 1/n + \alpha' n^{(\delta_a - 2)}$ ,  $\beta = 1 - 1/n + \beta' n^{(\delta_b - 2)}$ ,  $\gamma = 1 - 1/n + \gamma' n^{(\delta_c - 2)}$  satisfying  $\alpha' \beta' n^{(\delta_a + \delta_b - 1)} + \gamma' n^{(\delta_c - 1)} = 1$  such that  $G$  does not have a triangle-factor. To see this, consider a tripartite graph  $G$  with parts  $A$ ,  $B$  and  $C$ . Denote the edge densities of  $G[A, B]$ ,  $G[A, C]$  and  $G[B, C]$  by  $\alpha$ ,  $\beta$  and  $\gamma$ , respectively. Let  $x \in A$ ,  $B_1$  be a subset of  $B$  such that  $|B_1| = \alpha n^2 - n(n-1)$  and  $C_1$  be a subset of  $C$  such that  $|C_1| = \beta^2 - n(n-1)$ . Define the edge set of  $G$  to be the set of all edges between  $A \setminus \{x\}$  and  $B$ ,  $A \setminus \{x\}$  and  $C$ ,  $\{x\}$  and  $B_1$ ,  $\{x\}$  and  $C_1$ .

The rest of the paper is organized as follows. In Section 2, we will introduce notions and calculations needed for our proofs. Section 3 is devoted to prove our main result. In section 4, we will prove the triangle-factor case.

## 2 Preliminaries

Let

$$R = \{(\alpha, \beta, \gamma) \in [0, 1]^3 : \alpha\beta + \gamma > 1, \alpha\gamma + \beta > 1, \beta\gamma + \alpha > 1\},$$

and

$$\Delta(\alpha, \beta, \gamma) = \alpha^2 + \beta^2 + \gamma^2 - 2\alpha\beta - 2\alpha\gamma - 2\beta\gamma + 4\alpha\beta\gamma.$$

$R_1$  and  $R_2$  is a partition of  $R$  such that

$$R_1 = \{(\alpha, \beta, \gamma) \in R : \Delta(\alpha, \beta, \gamma) \geq 0\}$$

and  $R_2 = R \setminus R_1$ .

Let  $T_{\min}(\alpha, \beta, \gamma)$  denote the minimum number of triangles in tripartite graph with the stated densities, divided by  $n^3$ . From [BJT10], we have the following:

**Theorem 2.1** ( Baber, Johnson and Talbot [BJT10]). *If  $(\alpha, \beta, \gamma) \in R_1$ , then  $T_{\min}(\alpha, \beta, \gamma) = \alpha + \beta + \gamma - 2$ . If  $(\alpha, \beta, \gamma) \in R_2$ , then  $T_{\min}(\alpha, \beta, \gamma) \leq 2\sqrt{\alpha\beta(1-\gamma)} + 2\gamma - 2$ .*

**Conjecture 2.1** ( Baber, Johnson and Talbot [BJT10]). *If  $(\alpha, \beta, \gamma) \in R_2$ , then  $T_{\min}(\alpha, \beta, \gamma) = 2\sqrt{\alpha\beta(1-\gamma)} + 2\gamma - 2$ .*

For integers  $n > r \geq 1$ , let  $[n] := \{1, 2, \dots, n\}$  be the standard  $n$  element set and  $\binom{[n]}{r} = \{T \subseteq [n] : |T| = r\}$  be the collection of all its  $r$ -subsets. An  $n$ -vertex  $r$ -uniform hypergraph  $H$  is a pair  $H = (V, E)$ , where  $V := [n]$  and  $E(H) \subset \binom{[n]}{r}$ . Let  $\nu(H)$  be the *matching number* of  $H$ , that is, the maximum number of pairwise vertex-disjoint members of  $E(H)$ . An  $r$ -uniform hypergraph  $H$  is called  *$n$ -balanced  $r$ -partite* if  $V(H)$  is partitioned into sets  $V_1, V_2, \dots, V_r$  such that  $|V_1| = |V_2| = \dots = |V_r| = n$  and each hyperedge meets every  $V_i$  in precisely one vertex. Aharoni and Howard [AH17] showed the tight hyperedge density needed to guarantee the existence of a matching of size  $k$  in an  $n$ -balanced  $r$ -partite  $r$ -uniform hypergraphs. (see Observation 1.9 in [AH17])

**Lemma 2.2** (Aharoni and Howard [AH17]). *If  $H$  is an  $n$ -balanced  $r$ -partite  $r$ -uniform hypergraph and  $e(H) > (k-1)n^{r-1}$  then  $\nu(H) \geq k$  and the bound is tight.*

For a graph  $G$  and  $U \subseteq V(G)$ , we use  $G - U$  to denote the subgraph of  $G$  induced by  $V(G) \setminus U$ , and we use  $G[U]$  to denote the subgraph of  $G$  induced by  $U$ . For a graph  $G$  and  $E \subseteq E(G)$ , we use  $G - E$  to denote the graph obtained from  $G$  by deleting  $E$ .